



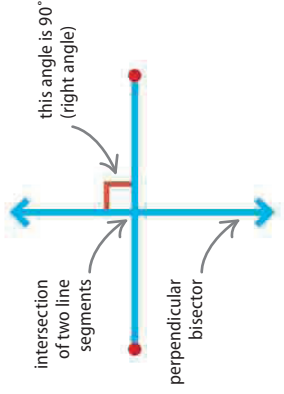
Constructions

MAKING PERPENDICULAR LINES AND ANGLES USING A COMPASS AND A STRAIGHT EDGE.

An accurate geometric drawing is called a **construction**. These drawings can include **line segments**, **angles**, and **shapes**. The tools needed are a **compass** and a **straight edge**.

Constructing perpendicular lines

Two line segments are perpendicular when they intersect (or cross) at 90° , or right angles. There are two ways to construct a perpendicular line—the first is to draw through a point marked on a given line segment; the second is to use a point above or below the segment.



◁ **Perpendicular bisector**
A perpendicular bisector cuts another line segment exactly in half, crossing through its midpoint at right angles, or 90° .

SEE ALSO

◀ **82–83** Tools in geometry

◀ **84–85** Angles

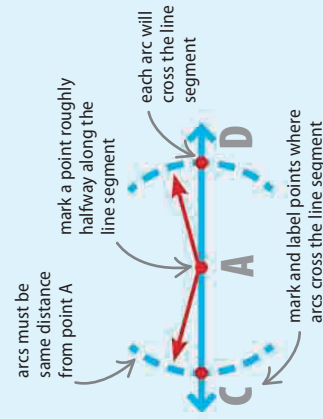
116–117 ▶

Congruent triangles

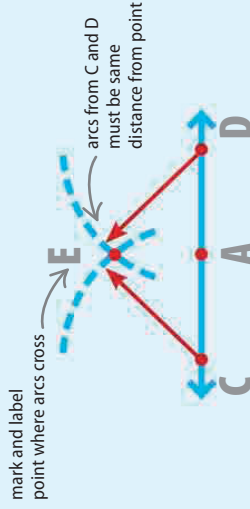
120–121 ▶

Using a point on the line segment

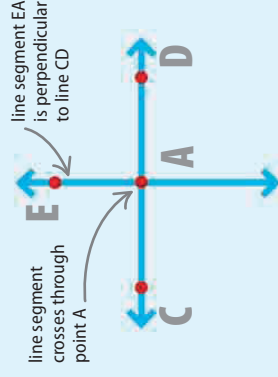
A perpendicular line can be constructed using a point marked on a line segment. The point marked is where the two lines will intersect (cross) at right angles.



Draw a line segment and mark a point on the segment with a letter, for example, A. Place the point of a compass on point A, and draw two arcs of the same distance from this point.



Place the point of a compass on point C, and draw an arc above the line segment. Do the same from point D. The arcs intersect (cross) at a point, label this point E.



Now, draw a line segment from E through A. This line segment is perpendicular (at right angles) to the original one.